

Explicit structure of the solution with polynomial tail to the Boltzmann equation

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Outline

Boltzmann Equation with linear scattering forcing term

Decomposition of the solution with polynomial tail

Iteration scheme

Boltzmann with polynomial tail

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Boltzmann Equation with linear scattering forcing term

$$\begin{cases} \partial_t F + \xi \cdot \nabla_x F = Q(F, F) + Q(\mathcal{M}, F) \\ F|_{t=0} = F_0(x, \xi), \end{cases}$$

t : time; $x \in \mathbb{R}^3$: space; $\xi \in \mathbb{R}^3$: **microscopic velocity**.

$F(t, x, \xi)$: density distribution function.

- **Transport:** $\partial_t F + \xi \cdot \nabla_x F$.

- **Collision operator: Hard sphere**

$$Q(G, F) = \int_{\mathbb{R}^3} \int_{S^2} |(\xi - \xi_*) \cdot n| [G(\xi'_*) F(\xi') - G(\xi_*) F(\xi)] dnd\xi_*$$

- ξ, ξ_* : velocity before collision, ξ', ξ'_* : velocity after collision.

$$\xi' = \xi - [(\xi - \xi_*) \cdot n]n, \quad \xi'_* = \xi_* + [(\xi - \xi_*) \cdot n]n.$$

- Scattering forcing: $Q(\mathcal{M}, F)$, $\mathcal{M} = \frac{1}{(2\pi)^{3/2}} \exp\left(-\frac{|\xi|^2}{2}\right)$.

Perturbation Problem

- $Q(F, F) + Q(\mathcal{M}, F) = 0 \Leftrightarrow F = \text{span}\{\mathcal{M}\}$
- Around background Maxwellian with large tail:

$$F = \mathcal{M} + f, F_0(x, \xi) = \mathcal{M} + \varepsilon f_0,$$

$$\begin{cases} \partial_t f + \xi \cdot \nabla_x f = \mathcal{L}f + Q(f, f) \\ f|_{t=0} = F_0(x, \xi) - \mathcal{M} = \varepsilon f_0(x, \xi) \end{cases}$$

$$\mathcal{L}f := [Q(\mathcal{M}, f) + Q(f, \mathcal{M})] + Q(\mathcal{M}, f) = -\nu(\xi)f + \mathcal{K}f.$$

- $\nu(\xi) \sim 1 + |\xi|$, $\mathcal{K}f$: integral operator
- $f_0 \in L_{\xi, \beta}^\infty L_x^\infty$ with $\beta > 4$, compactly supported in x for all ξ :

$$f_0(x, \xi) \equiv 0 \quad \text{for } |x| \geq 1, \xi \in \mathbb{R}^3$$

- **Goal: Pointwise space-time structure of f**

Review of previous work

Boltzmann equation with polynomial tail

- Torus: Gualdani-Mischler-Mouhot (2017)
- Whole space: Duan-Li-Liu (2024), Carrapatoso-Gervais (2024)

Explicit structure of the kinetic and fluid equations

- Boltzmann with hard sphere (Gaussian tail). 1D: Liu-Yu (2004), 3D linear: Liu-Yu (2006, 2011)
- Compressible NS. 1D: Liu-Zeng (1997), 3D: Liu-Wang (1998), Liu-Noh (2015), Deng-Yu (2017), Du-Wu (2017)

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Decomposition

We introduce the decomposition $f = f_1 + \sqrt{\mathcal{M}}f_2$ with

$$\begin{cases} \partial_t f_1 + \xi \cdot \nabla_x f_1 = \nu(\xi) f_1 + \mathcal{K}_s f_1 + Q(f_1 + \sqrt{\mathcal{M}}f_2, f_1 + \sqrt{\mathcal{M}}f_2) \\ \partial_t f_2 + \xi \cdot \nabla_x f_2 = Lf_2 + K_b f_1 \end{cases}$$

with initial data

$$f_1(0, x, \xi) = \varepsilon f_0(x, \xi), \quad f_2(0, x, \xi) = 0,$$

where

$$Lg = \frac{1}{\sqrt{\mathcal{M}}} \left[Q(\mathcal{M}, \sqrt{\mathcal{M}}g) + Q(\sqrt{\mathcal{M}}g, \mathcal{M}) + Q(\mathcal{M}, \sqrt{\mathcal{M}}g) \right]$$

$$\mathcal{K}_s := \chi_{\{|v| \geq M\}} \mathcal{K}, \quad \mathcal{K}_b := \chi_{\{|v| < M\}} \mathcal{K}, \quad K_b = \mathcal{M}^{-1/2} \mathcal{K}_b$$

for a constant $M > 0$ large enough, $\chi_{\{\cdot\}}$ being the indicator function.

Decomposition

- **Behavior of L :** $\ker(L) = \text{span}\{\sqrt{\mathcal{M}}\}$. Let P_0 be the orthogonal projection with respect to the L_ξ^2 inner product onto $\ker(L)$, and $P_1 \equiv \text{Id} - P_0$.

- **Behavior of $\mathcal{K}_s$:** For any $\beta > 4$, there is a constant $C_\beta > 0$ such that for any $M > 0$,

$$\chi_{\{|\xi| \geq M\}} \langle \xi \rangle^\beta |\mathcal{K}f| \leq \left(\frac{C}{\beta} + \frac{C_\beta}{M^2} \right) \nu(\xi) |f|_{L_{\xi,\beta}^\infty}$$

- **Behavior of K_b :** For $k > 4$ and any $j \geq 0$, then for $1 \leq p \leq \infty$,

$$\|K_b f\|_{L_{\xi,j}^\infty L_x^p} \leq C(j, k) \|f\|_{L_{\xi,k}^\infty L_x^p}$$

- **Behavior of Q :** $\left| \langle \xi \rangle^\beta Q(f, g) \right| \leq C_\beta \nu(\xi) |f|_{L_{\xi,\beta}^\infty} |g|_{L_{\xi,\beta}^\infty}$

Green's function

- $\mathbb{S}^t$: $\partial_t g + \xi \cdot \nabla_x g + \nu(\xi) g = 0$
- $\tilde{\mathbb{S}}^t$: $\partial_t g + \xi \cdot \nabla_x g + \nu(\xi) g = \mathcal{K}_s g$
- $\mathbb{G}^t$: $\partial_t g + \xi \cdot \nabla_x g = Lg$

If $g_0 \in L_{\xi, \beta}^\infty L_x^\infty$ is compactly supported in x :

$$g_0(x, \xi) \equiv 0 \quad \text{for } |x| \geq 1, \xi \in \mathbb{R}^3$$

- $\left| \tilde{\mathbb{S}}^t g_0 \right|_{L_{\xi, \beta}^\infty} \leq \mathfrak{C} e^{-\frac{\nu_0}{3}(t+|x|)} \|g_0\|_{L_{\xi, \beta}^\infty L_x^\infty}, \beta > 4$ sufficiently large, constant $\mathfrak{C} > 0$ independent of M and β .

- $|\mathbb{G}^t g_0|_{L_{\xi, \beta}^\infty} \leq \mathfrak{C}_1 \left[(1+t)^{-3/2} e^{-\frac{|x|^2}{D_0(1+t)}} + e^{-\frac{t+|x|}{c_0}} \right] \|g_0\|_{L_{\xi, \beta}^\infty L_x^\infty},$ for any $\beta > 3/2$ and some constants $\mathfrak{C}_1, c_0, D_0 > 0$.

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Iteration scheme

- $f^{(n)} = f_1^{(n)} + \sqrt{\mathcal{M}} f_2^{(n)}$:

$$\begin{cases} \partial_t f_1^{(n+1)} + \xi \cdot \nabla_x f_1^{(n+1)} + \nu(\xi) f_1^{(n+1)} = \mathcal{K}_s f_1^{(n+1)} + Q(f^{(n)}, f^{(n)}) \\ \partial_t f_2^{(n+1)} + \xi \cdot \nabla_x f_2^{(n+1)} = L f_2^{(n+1)} + K_b f_1^{(n+1)} \end{cases}$$

with $f_1^{(n+1)}(0, x, \xi) = \varepsilon f_0(x, \xi)$, $f_2^{(n+1)}(0, x, \xi) = 0$

- Initial step: $f_1^{(0)}(t, x, \xi) = f_2^{(0)}(t, x, \xi) = 0$

- $\left| f_1^{(1)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} e^{-\frac{\nu_0}{3}(t+|x|)}$

- $f_2^{(1)} = \int_0^t \mathbb{G}^{t-\tau} K_b f_1^{(1)}(\tau, x, \xi) d\tau$

- $\left| f_2^{(1)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim$

$$\varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} \left[(1+t)^{-3/2} e^{-\frac{|x|^2}{D_0(1+t)}} + e^{-\frac{t+|x|}{c_0}} \right] *_{t,x} e^{-\frac{\nu_0}{3}(t+|x|)}$$

Estimate of $f_2^{(1)}$

- Lemma:** Let D, σ, ϖ be positive constants, $\alpha \geq 0$. Then for any $1/2 < \theta, \lambda < 1$,

$$\int_0^t \int_{\mathbb{R}^3} \frac{e^{-\frac{|x-y|^2}{D(1+t-s)}}}{(1+t-s)^{\alpha/2}} e^{-\frac{2\sigma|y|+\varpi s}{D}} dy ds$$

$$\leq \begin{cases} \tilde{C}_\theta \left(e^{-\theta \frac{2\sigma|x|+\varpi t}{D}} + (1+t)^{-\alpha/2} e^{-\frac{1}{C_\theta} \frac{|x|^2}{D(1+t)}} \right) \\ \tilde{C}_\lambda \left((1+t)^{-\alpha/2} e^{-\lambda \frac{|x|^2}{D(1+t)}} + e^{-\frac{1}{C_\lambda} \frac{2\sigma|x|+\varpi t}{D}} \right) \end{cases}, \quad t \geq 0,$$

where $C_\theta = C(\theta, \sigma, \varpi)$, $C_\lambda = C(\lambda, \sigma, \varpi) > 0$.

- $\left| f_2^{(1)} \right|_{L_{\xi, \beta}^\infty} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^\infty L_x^\infty} \left[(1+t)^{-3/2} e^{-\frac{|x|^2}{2\hat{D}(1+t)}} + e^{-\frac{t+|x|}{2\hat{c}}} \right], \text{ for } \hat{c}$

and $\hat{D}$ sufficiently large.

Estimate of $f_1^{(2)}$

- $f_1^{(2)} = h_1^{(2)} + (h_2^{(2)} + h_3^{(2)})$: initial+source
- $Q(f_1^{(1)} + \sqrt{\mathcal{M}}f_2^{(1)}, f_1^{(1)} + \sqrt{\mathcal{M}}f_2^{(1)}) = U_2^{(1)} + U_3^{(1)}$
 $U_2^{(1)} = Q(f_1^{(1)}, f_1^{(1)}) + Q(f_1^{(1)}, \sqrt{\mathcal{M}}f_2^{(1)}) + Q(\sqrt{\mathcal{M}}f_2^{(1)}, f_1^{(1)})$
 $U_3^{(1)} = Q(\sqrt{\mathcal{M}}f_2^{(1)}, \sqrt{\mathcal{M}}f_2^{(1)})$: **gain extra velocity decay**

$$\begin{cases} \partial_t h_1^{(2)} + \xi \cdot \nabla_x h_1^{(2)} + \nu(\xi) h_1^{(2)} = \mathcal{K}_s h_1^{(2)} \\ h_1^{(2)}(0, x, \xi) = \varepsilon f_0(x, \xi), \end{cases}$$

For $i = 2, 3$

$$\begin{cases} \partial_t h_i^{(2)} + \xi \cdot \nabla_x h_i^{(2)} + \nu(\xi) h_i^{(2)} = \mathcal{K}_s h_i^{(2)} + U_i^{(1)} \\ h_i^{(2)}(0, x, \xi) = 0. \end{cases}$$

Estimate of $f_1^{(2)}$

- $\left| h_1^{(2)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} e^{-\frac{\nu_0}{3}(t+|x|)}$
- $\left| h_2^{(2)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} e^{-\frac{1}{2c_2}(t+|x|)} \text{ (Using } \mathbb{S}^t \text{)}$
- $\left| h_3^{(2)} \right|_{L_{\xi, \beta+m}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} \left[\frac{e^{-\frac{|x|^2}{2D_3(1+t)}}}{(1+t)^3} + e^{-\frac{t+|x|}{2c_3}} \right] \text{ (Using } \tilde{\mathbb{S}}^t \text{)}$

for some c_2, c_3, D_3 large

- $f^{(n)} = f_1^{(n)} + \sqrt{\mathcal{M}}f_2^{(n)}, f_1^{(n)} = h_1^{(n)} + h_2^{(n)} + h_3^{(n)}$

$$\begin{cases} \partial_t h_1^{(n)} + \xi \cdot \nabla_x h_1^{(n)} + \nu(\xi) h_1^{(n)} = \mathcal{K}_s h_1^{(n)} \\ h_1^{(n)}(0, x, \xi) = \varepsilon f_0(x, \xi) \end{cases}$$

For $i = 2, 3$,

$$\begin{cases} \partial_t h_i^{(n)} + \xi \cdot \nabla_x h_i^{(n)} + \nu(\xi) h_i^{(n)} = \mathcal{K}_s h_i^{(n)} + U_i^{(n-1)}(t, x, \xi) \\ h_i^{(n)}(0, x, \xi) = 0 \end{cases}$$

- $U_2^{(n-1)} =$
 $Q\left(h_1^{(n-1)} + h_2^{(n-1)}, h_1^{(n-1)} + h_2^{(n-1)} + h_3^{(n-1)} + \sqrt{\mathcal{M}}f_2^{(n-1)}\right)$
 $+ Q\left(h_3^{(n-1)} + \sqrt{\mathcal{M}}f_2^{(n-1)}, h_1^{(n-1)} + h_2^{(n-1)}\right)$

- $U_3^{(n-1)} = Q\left(h_3^{(n-1)} + \sqrt{\mathcal{M}}f_2^{(n-1)}, h_3^{(n-1)} + \sqrt{\mathcal{M}}f_2^{(n-1)}\right)$

Ansatz

- $h_1^{(n)}$: initial problem (fixed)

$$\left| h_1^{(n)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} e^{-\frac{\nu_0}{3}(t+|x|)}$$

- $h_2^{(n)}$: source **lost** velocity decay

$$\left| h_2^{(n)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} e^{-\frac{1}{2c_2}(t+|x|)}$$

- $h_3^{(n)}$: source **with** extra velocity decay

$$\left| h_3^{(n)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} \left[(1+t)^{-3} e^{-\frac{|x|^2}{2D_3(1+t)}} + e^{-\frac{t+|x|}{2c_3}} \right]$$

- $\left| f_2^{(n)} \right|_{L_{\xi, \beta}^{\infty}} \lesssim \varepsilon \|f_0\|_{L_{\xi, \beta}^{\infty} L_x^{\infty}} \left(e^{-\frac{1}{2\hat{c}}(t+|x|)} + (1+t)^{-3/2} e^{-\frac{|x|^2}{2\hat{D}(1+t)}} \right)$

- $c_2, c_3, D_3, \hat{c}, \hat{D}$ to be determined

Conclusion

- $F = \mathcal{M} + f$ with $f = f_1 + \sqrt{\mathcal{M}}f_2$, then

$$|f_1|_{L_{\xi,\beta}^\infty} \lesssim \varepsilon \|f_0\|_{L_{\xi,\beta}^\infty L_x^\infty} \left((1+t)^{-3} e^{-\frac{|x|^2}{2D_3(1+t)}} + e^{-\frac{t+|x|}{2\hat{c}_3}} \right),$$

$$|f_2|_{L_{\xi,\beta}^\infty} \lesssim \varepsilon \|f_0\|_{L_{\xi,\beta}^\infty L_x^\infty} \left((1+t)^{-3/2} e^{-\frac{|x|^2}{2\hat{D}(1+t)}} + e^{-\frac{1}{2\hat{c}}(t+|x|)} \right).$$

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$$\partial_t F + \xi \cdot \nabla_x F = Q(F, F)$$

Write $F = \mathcal{M} + f$, and $f = f_1 + \sqrt{\mathcal{M}}f_2$:

$$\begin{cases} \partial_t f_1 + \xi \cdot \nabla_x f_1 = \nu(\xi) f_1 + \mathcal{K}_s f_1 + Q(f_1 + \sqrt{\mathcal{M}}f_2, f_1 + \sqrt{\mathcal{M}}f_2), \\ \partial_t f_2 + \xi \cdot \nabla_x f_2 = Lf_2 + K_b f_1, \end{cases}$$

where $Lg = \mathcal{M}^{-1/2} \left[Q(\mathcal{M}, \sqrt{\mathcal{M}}g) + Q(\sqrt{\mathcal{M}}g, \mathcal{M}) \right]$.

Use Green's function to write it into an integral system:

$$f_1 = \mathbb{S}^t f_0 + \int_0^t \mathbb{S}^{t-s} (\mathcal{K}_s f_1 + Q(f, f))(s) ds, \quad f_2 = \int_0^t \mathbb{G}^{t-s} K_b f_1(s) ds.$$

Compare with Gaussian tail perturbation $F = \mathcal{M} + \sqrt{\mathcal{M}}f$,

$$\partial_t f + \xi \cdot \nabla_x f = Lf + \Gamma(f, f), \quad f = \mathbb{G}^t f_0 + \int_0^t \mathbb{G}^{t-s} \Gamma(f, f)(s) ds.$$

Structure of $\mathbb{G}^t$

For initial data f_0 compact support in space with good velocity decay, Liu-Yu (2006, 2011) showed

$$\begin{aligned} |\mathbb{G}^t f_0|_{L_{\xi, \beta}^\infty} \leq C & \left[(1+t)^{-3/2} e^{-\frac{|x|^2}{D_0(1+t)}} + (1+t)^{-2} e^{-\frac{(|x| - ct)^2}{D_0(1+t)}} \right. \\ & \left. + \chi_{\{|x| \leq ct\}} (1+t)^{-3/2} \left(1 + \frac{|x|^2}{1+t} \right)^{-3/2} + e^{-\frac{t+|x|}{c_0}} \right] \end{aligned}$$

for any $\beta > 3/2$, here $c = \sqrt{5/3}$ is the sound speed associated with $\mathcal{M} = (2\pi)^{-3/2} e^{-|\xi|^2/2}$.

Moreover, if f_0 is purely microscopic, one gets extra $(1+t)^{-1/2}$ decay. $\Gamma(f, g)$ is purely microscopic.

Difficulty

$$f_2 = \int_0^t \mathbb{G}^{t-s} K_b f_1(s) ds, \quad f = \mathbb{G}^t f_0 + \int_0^t \mathbb{G}^{t-s} \Gamma(f, f)(s) ds$$

Concerning time-asymptotic behavior, the Gaussian tail case is similar to compressible N-S. The perturbation solution to CNS can be written symbolically as

$$u(t) = e^{tL} u_0 + \int_0^t e^{tL} \partial \mathcal{N}(u)(s) ds.$$

The time-asymptotic behavior of e^{tL} is similar as $\mathbb{G}^t$. Moving ∂ to semi-group e^{tL} **gains extra decay** $(1+t)^{-1/2}$. Previous studies on pointwise structure **ALL** rely on this property.

We **DO NOT** have it for polynomial tail. We have to get **sharp estimates for nonlinear interactions between slower decay waves**.

Result (still in progress)

- Ansatz (suitable for inside the cone):

$$|f_1|_{L_{\xi,\beta}^\infty} \lesssim \varepsilon \left[(1+t)^{-3} \left(1 + \frac{|x|^2}{1+t}\right)^{-3} + (1+t)^{-4} \left(1 + \frac{(|x| - ct)^2}{1+t}\right)^{-2} \right]$$

$$|f_2|_{L_{\xi,\beta}^\infty} \lesssim \varepsilon \left[(1+t)^{-3/2} \left(1 + \frac{|x|^2}{1+t}\right)^{-3/2} + (1+t)^{-2} \left(1 + \frac{(|x| - ct)^2}{1+t}\right)^{-1} \right]$$

- Weighted energy method for outside the cone: $w = \exp(\frac{\langle x \rangle - Mt}{D})$

and let $u = wf = u_1 + \sqrt{\mathcal{M}}u_2$

- Wave-remainder decomposition + energy estimate

- Conclusion: If $M > c$

- For $|x| < Mt$, fluid structure

- For $|x| > Mt$, space time exponential decay

- How to find out the separation of scale structure for polynomial tail? i.e. Faster decay for microscopic data.

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Shih-Hsien, Happy Birth Day!